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2.1: Find laplace: $f(t) = te^{-5t}$

$\mathcal{L}\{e^{-at}f(t)\} = F(a+s)$ | $\mathcal{L}\{e^{-5t}t\} = F(s+5)$

$\mathcal{L}\{t\} = \frac{1}{s^2}$ | $F(s+5) = \frac{1}{(s+5)^2}$

2.2. Find Inverse laplace: $F(s) = \frac{10}{s(s+2)(s+3)^2}$

$$\frac{10}{s(s+2)(s+3)^2} = \frac{K_1}{s} + \frac{K_2}{s+2} + \frac{K_3}{(s+3)^2} + \frac{K_4}{s+3}$$

K_1: $\frac{10s}{s(s+2)(s+3)^2} + \frac{K_1 s}{s} + \frac{K_2}{s+2}s + \frac{K_3}{(s+3)^2}s + \frac{K_4}{(s+3)}s$ | $s \to 0$

$\frac{10}{2(3)^2} = \boxed{\frac{10}{18} = K_1 = \frac{5}{9}}$

K_2: $\frac{10}{s(s+3)^2} = \frac{K_1}{s}(s+2) + K_2 + \frac{K_3}{(s+3)^2}(s+2) + \frac{K_4}{s+3}(s+2)$ | $s \to -2$

$\frac{10}{-2(+1^2)} = \boxed{-5 = K_2}$

K_3: $\frac{10}{s(s+2)} = \frac{K_1}{s}(s+3)^2 + \frac{K_2}{s}(s+3)^2 + K^3 + \frac{K_4}{(s+3)}(s+3)^2$ | $s \to -3$

$\frac{10}{s^2+2s}$

$\frac{10}{-3(-1)} = \boxed{\frac{10}{3} = K_3}$

K_4: $\frac{-10(2s+2)}{(s^2+2s)^2} = K_4$ | $s \to -3$

$\frac{-10(-4)}{(9-6)^2} = \boxed{\frac{40}{9} = K_4}$

$$F(s) = \frac{10}{s(s+2)(s+3)^2} = \frac{5}{9}\left(\frac{1}{s}\right) + -5\left(\frac{1}{s+2}\right) + \frac{10}{3}\left(\frac{1}{(s+3)^2}\right) + \frac{40}{9}\left(\frac{1}{s+3}\right)$$

$$\mathcal{L}^{-1}\{F(s)\} = \boxed{\frac{5}{9} - 5e^{-2t} + \frac{10}{3}te^{-3t} + \frac{40}{9}e^{-3t}}$$

$$f(t) = \left(\frac{5}{9} - 5e^{-2t} + \frac{10}{3}te^{-3t} + \frac{40}{9}e^{-3t}\right)u(t)$$

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2.3 find transfer func. $G(s) = C(s) / R(s)$

$$\frac{d^3c}{dt^3} + 3\frac{d^2c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2r}{dt^2} + 4\frac{dr}{dt} + 3r$$

$$s^3C(s) + 3s^2C(s) + 7sC(s) + 5C(s) = s^2R(s) + 4sR(s) + 3R(s)$$

$$(s^3 + 3s^2 + 7s + 5)C(s) = (s^2 + 4s + 3)R(s)$$

$$\frac{C(s)}{R(s)} = \frac{s^2 + 4s + 3}{s^3 + 3s^2 + 7s + 5}$$

2.4. find differential of transfer func.

$$G(s) = \frac{2s + 1}{s^2 + 6s + 2} = \frac{C(s)}{R(s)}$$

$$\mathcal{L}^{-1}\{(2s + 1)(R(s)) = (s^2 + 6s + 2)C(s)\}$$

$$2\frac{dr}{dt} + r = \frac{d^2c}{dt^2} + 6\frac{dc}{dt} + 2c$$

2.5 Find ramp response in: $G(s) = \frac{s}{(s+4)(s+8)}$

$r(t) = t \cdot u(t) =$
$\mathcal{L}\{r(t)\} = \frac{1}{s^2}$ $\quad G(s) = \frac{C(s)}{R(s)} \therefore C(s) = G(s)R(s) = \frac{s}{(s+4)(s+8)}\left(\frac{1}{s^2}\right) = \frac{1}{s(s+4)(s+8)}$

$$\frac{1}{s(s+4)(s+8)} = \frac{K_1}{s} + \frac{K_2}{s+4} + \frac{K_3}{s+8}$$

K_1: $\frac{1}{(s+4)(s+8)} = K_1$; $s \to 0$ $\quad \frac{1}{4(8)} = \boxed{\frac{1}{32} = K_1}$

K_2: $\frac{1}{s(s+8)} = K_2$; $s \to -4$ $\quad \frac{1}{-4(4)} = \boxed{\frac{1}{-16} = K_2}$

K_3: $\frac{1}{s(s+4)} = K_3$; $s \to -8$ $\quad \frac{1}{-8(-4)} = \boxed{\frac{1}{32} = K_3}$

$$C(s) = \frac{1}{32}\left(\frac{1}{s}\right) + \frac{-1}{16}\left(\frac{1}{(s+4)}\right) + \frac{1}{32}\left(\frac{1}{s+8}\right)$$

$$\mathcal{L}^{-1}\{C(s)\} = \frac{1}{32} - \frac{1}{16}e^{-4t} + \frac{1}{32}e^{-8t}$$

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